CANONICAL MORPHISM FROM PROJ(S) TO SPEC(S_0)

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Morphism into affine. schemes-lemma-morphism-into-affine (cross-chapter, named). Produces the morphism from the S_0-algebra structure on the structure sheaf.
- Standard open algebra maps. lemma-standard-open (carded this batch). S to S_f and S_(f) to S_f are S_0-algebra maps.

Source reference. Original reference 00JN: focused-seeds/geometry/S9-cons-structure-morphism-proj/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01M4: sources/stacks-constructions/pinned-original-source.tex, source lines 1172–1172. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01I1: focused-seeds/geometry/S9-cons-structure-morphism-proj/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. The scheme $\mathit{Proj}(S)$ has a canonical morphism towards the affine scheme $\mathrm{Spec}(S_0)$, agreeing with the map on topological spaces coming from Algebra, Definition 00JN.*

Proof. We saw above that our construction of $\widetilde{S}$, resp. $\widetilde{M}$ gives a sheaf of S_0-algebras, resp. S_0-modules. Hence we get a morphism by Schemes, Lemma 01I1. This morphism, when restricted to $D_+(f)$ comes from the canonical ring map $S_0 \to S_{(f)}$. The maps $S \to S_f$, $S_{(f)} \to S_f$ are S_0-algebra maps, see Lemma 01M4. Hence if the homogeneous prime $\mathfrak{p} \subset S$ corresponds to the $\mathbf{Z}$-graded prime $\mathfrak{p}' \subset S_f$ and the (usual) prime $\mathfrak{p}'' \subset S_{(f)}$, then each of these has the same inverse image in S_0. $\square$

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.